



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

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Executive Summary

Summary of methodologies

Data Collection

Data Wrangling

Predictive Modeling

Exploratory Data Analysis

Summary of all results

- This project focuses on analyzing SpaceX Falcon 9 launch data to predict rocket landing success rates, leveraging historical data collected via API calls and web scraping. The data was cleaned, transformed, and explored using both SQL queries and visualization techniques, followed by the implementation of various machine learning models to make predictions.
- This analysis revealed key factors influencing the success of Falcon 9 landings, such as payload mass, launch site, and orbit type. The best-performing model, the Decision Tree, provides SpaceX with actionable insights to improve landing protocols and mission planning, ultimately contributing to reducing costs and enhancing reusability.
- This project successfully demonstrated the use of data analytics and predictive modeling to optimize future SpaceX missions by providing a framework for more accurate landing predictions.

Introduction

Project background and context

SpaceX, a private aerospace manufacturer and space transportation company, has revolutionized the space industry by successfully developing reusable rockets. One of their key goals is to reduce the cost of space travel by landing rocket boosters after launches. This project focuses on analyzing historical launch data to predict the success of these rocket landings.

The project utilizes data from SpaceX's Falcon 9 launches, which are retrieved through API calls and web scraping techniques. By analyzing this data, the goal is to identify patterns and key factors that influence whether a Falcon 9 rocket will successfully land or not.

Problems you want to find answers

The primary problem addressed in this project is to predict whether SpaceX will attempt to land a Falcon 9 rocket successfully. SpaceX's ability to consistently land rockets is critical to reducing costs and enhancing the reusability of spacecraft. The prediction involves assessing various launch characteristics, such as rocket type, payload weight, and landing pad usage, and determining their impact on the likelihood of a successful landing.

This problem was motivated by the need to optimize rocket landings and improve the decision-making process for future missions. By understanding the key drivers of landing success, SpaceX can fine-tune their technology and launch protocols to ensure higher success rates and reduce mission costs.

Section 1

Methodology

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Methodology

Executive Summary

1. Data Collection Methodology

The SpaceX Falcon 9 launch data was collected using two main techniques:

API Data Collection:

Data was gathered using the SpaceX REST API (api.spacexdata.com/v4/launches/past). This provided detailed information about the launches, rockets, payloads, and landing outcomes. The data was extracted in JSON format and processed using the requests and Json libraries in Python.

Web Scraping:

Additional data was collected by web scraping HTML tables on Wikipedia that contain historical launch records of Falcon 9 missions. The BeautifulSoup library was employed to parse and extract the necessary data into a structured format.

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2. Data Wrangling

Once the raw data was collected, several data cleaning and transformation steps were applied:

Filtering Irrelevant Records:

The dataset included information on both Falcon 1 and Falcon 9 launches, but only Falcon 9 data was relevant. Falcon 1 launches were filtered out.

Handling Missing Values:

Certain columns, such as PayloadMass, had missing values. These were replaced with the mean of the available values to ensure consistency across the dataset.

Data Normalization:

Complex, nested JSON structures obtained from the API were flattened using the `json_normalize()` function to create a more usable tabular format for analysis.

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3. Exploratory Data Analysis (EDA)

Exploratory data analysis was conducted to uncover insights about the dataset:

SQL Queries:

SQL queries were executed on the dataset to analyze trends, such as the success rates of rocket landings over time, and the relationship between payload weight and landing outcomes.

Visualization:

Using libraries like matplotlib and seaborn, visualizations such as pie charts and scatter plots were created to identify key trends and patterns. These visuals helped understand the distribution of launches across various categories, such as launch sites and payload masses.

Methodology

Executive Summary

4. Interactive Visual Analytics

Interactive dashboards were built to facilitate further exploration:

Folium Maps:

Interactive maps were created using the folium library, showcasing the launch sites, landing locations, and geographical relationships between them. The MousePosition feature was used to retrieve precise coordinates for visualization purposes.

Plotly Dash:

An interactive dashboard was developed using the Plotly Dash framework to allow users to visualize launch statistics, including payload ranges and success rates. Users could adjust filters (such as payload range) to dynamically update the visualizations.

Methodology

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5. Predictive Analysis: Classification Models

Predictive modeling was the key focus of this project. Several classification models were tested to predict whether SpaceX would successfully land a Falcon 9 rocket:

Model Selection:

Logistic Regression, Support Vector Machines (SVM), Decision Trees, and K-Nearest Neighbors (KNN) were selected as potential models.

Model Tuning:

Hyperparameter tuning was performed using GridSearchCV to optimize each model's performance. Parameters such as the regularization strength for Logistic Regression and the kernel type for SVM were adjusted to achieve better results.

Evaluation:

The models were evaluated based on metrics such as accuracy, precision, recall, and F1-score. The results of the models were compared to determine which one had the highest performance 10 on the test dataset.

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Executive Summary

6. Building, Tuning, and Evaluating Classification Models

Building Models:

After data preprocessing, the classification models were built using Python's scikit-learn library. The training data was used to fit the models, and they were evaluated using cross-validation techniques.

Tuning Models:

Hyperparameter tuning was performed to improve model performance. For instance, parameters such as the C value in Logistic Regression or the maximum depth in Decision Trees were optimized using GridSearchCV.

Evaluating Models:

The final models were evaluated based on their performance on unseen test data. Accuracy and confusion matrices were used to assess the overall model effectiveness, while precision, recall, and F1-scores provided deeper insights into the models' prediction capabilities.

Data Collection

The SpaceX Falcon 9 launch data was collected from two primary sources: the SpaceX REST API and web scraping from Wikipedia. Each method allowed us to gather comprehensive data on the launches, payloads, rockets, landing outcomes, and more. The following steps were involved in the data collection process:

1. Collecting Data from SpaceX REST API

Endpoint: We used the endpoint `api.spacexdata.com/v4/launches/past` to retrieve all past launch records.

Data Format: The data was obtained in JSON format, with each JSON object representing a single launch. This JSON data included details such as rocket ID, payload information, launch date, and success/failure status.

Steps:

- We used the requests library in Python to perform a GET request to the API endpoint.
- The response was converted into a Python dictionary using the `.json()` method.
- Nested data in JSON (such as rocket and payload IDs) was normalized into a flat table using `json_normalize()` for easier analysis.

Data Collection

2. Web Scraping Falcon 9 Data from Wikipedia

Target Web Page: We scraped HTML tables from a Wikipedia page containing Falcon 9 launch records.

Libraries: Python's BeautifulSoup and requests libraries were used to access and parse the webpage's HTML.

Steps:

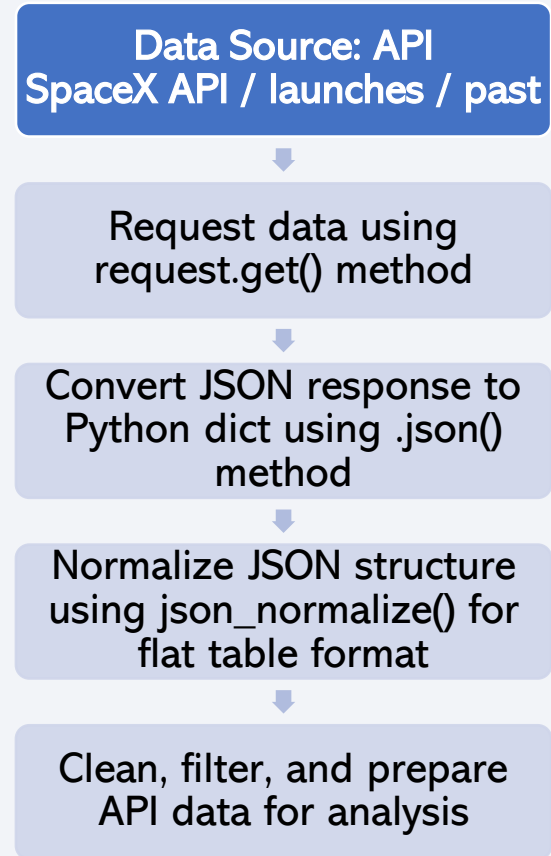
- We sent a request to the Wikipedia page using the requests library.
- Using BeautifulSoup, the HTML tables on the page were identified and parsed.
- Relevant columns, such as launch date, payload mass, and launch success, were extracted from the HTML tables and converted into a pandas DataFrame for further analysis.

Data Collection – SpaceX API

The GitHub URL of my completed
SpaceX API calls notebook:

[GitHub URL SpaceX API Calls](#)

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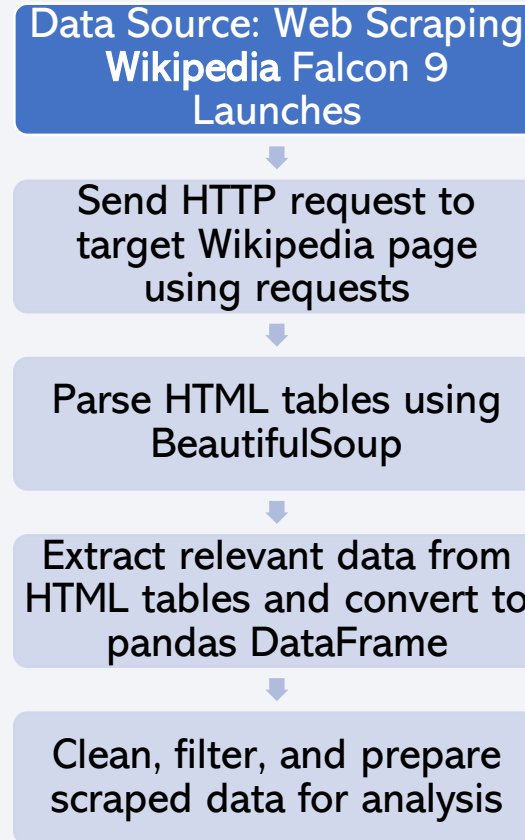


Data Collection - Scraping

The GitHub URL of my
completed web scraping
notebook:

[GitHub URL Webscraping](#)

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Data Wrangling

1. Data Cleaning

- **Handling Missing Values:**
 - The **PayloadMass** column contained missing values. We calculated the mean PayloadMass from the available data and replaced the missing values with this mean.
 - The **LandingPad** column also had missing values. Since the absence of a value in this column indicates that no landing pad was used, we left these values as NULL to be handled later in the modeling stage with one-hot encoding.
- **Data Filtering:** We filtered the dataset to remove data related to Falcon 1 launches, as our analysis focused solely on Falcon 9 rockets.

2. Data Transformation

- **API-Based Data Retrieval:** Certain columns, such as **rocket** and **payload**, contained unique identifiers instead of descriptive data. To address this:
 - We performed **additional API requests** targeting relevant endpoints to retrieve detailed information (e.g., rocket type, payload description) using these unique IDs.

Data Wrangling

- The resulting data was stored in lists and integrated into the dataset to replace the ID-based columns with more informative, human-readable data.

3. Dealing with Nested Data

- **JSON Normalization:** The API returned nested JSON structures. We used `json_normalize()` to flatten these nested structures, creating a tabular format suitable for analysis. For example:
 - Nested data about the rocket type and core serials were flattened into individual columns.

4. Feature Engineering

- Created **new features** from existing data to enhance predictive model performance, such as creating a categorical feature for **landing success** based on landing outcomes.

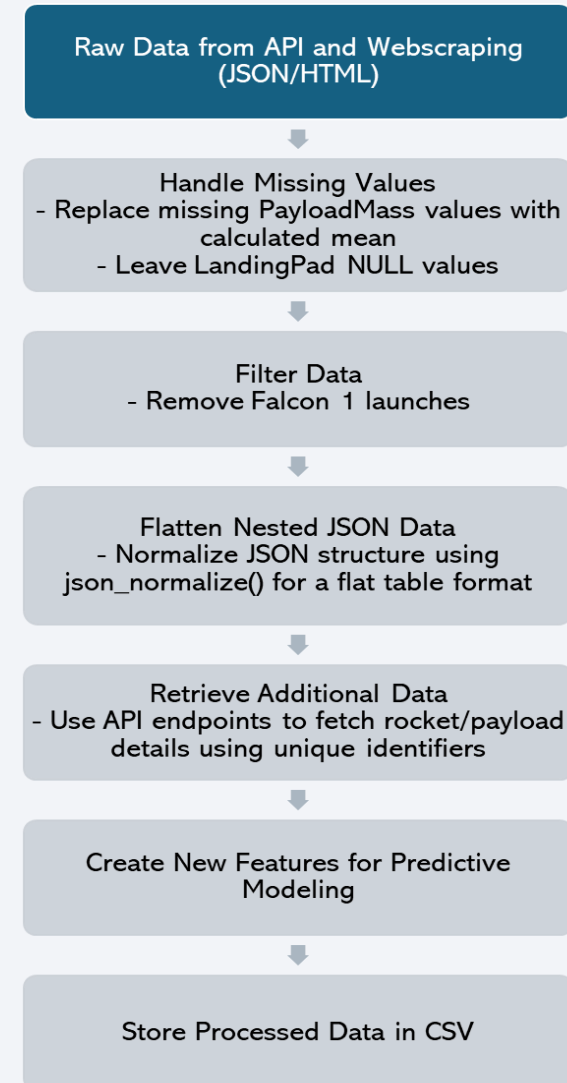
5. Storing the Processed Data

- After cleaning, transforming, and enhancing the dataset, we stored it in a structured format (CSV) to be used for further analysis and predictive modeling.

Data Wrangling

The GitHub URL of my completed
data wrangling related notebooks:
[GitHub URL Data Wrangling](#)

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EDA with Data Visualization

Chart Type: Scatter Plot

Purpose: This chart visualizes the relationship between **Flight Number and Launch Site**, **Payload Mass (kg) and Launch Site**, **Flight Number and Orbit Type** with the **class value** represented by color (hue). This helps us understand the frequency of launches at different sites and the success rate associated with each. This chart explores whether the payload mass has any impact on the success of launches at different sites. It helps analyze whether certain orbit types are more frequently associated with successful launches.

Insight: This visualization helps identify which launch sites have been used more frequently and how successful they have been over time. It indicates if heavier payloads are more successful at specific sites or if certain sites handle larger payloads more frequently. This chart provides insights into how frequently different orbit types are used and whether there is a relationship between flight number and the success of each orbit.

EDA with Data Visualization

Chart Type: Bar Chart

Purpose: The bar chart was plotted to compare the **success rate of each orbit type**. Each bar represents the proportion of successful launches for a specific orbit type, providing a clear comparison between them.

Insight: This visualization helps identify which orbit types have the highest or lowest success rates, providing insight into how orbit type may influence landing outcomes.

Chart Type: Line Chart

Purpose: This line chart visualizes the **yearly trend of launch success**, with the x-axis representing the year and the y-axis showing the success rate. The aim is to observe how SpaceX's launch success has improved over time.

Insight: This visualization provides a clear overview of SpaceX's improving launch success rate over the years, demonstrating the progress made in rocket reusability.

The GitHub URL of my completed EDA with data visualization notebook:

[GitHub URL EDA with Data Visualization](#)

EDA with SQL

- **Unique Launch Sites:** Retrieved a list of unique launch sites used in the missions.
- **Launch Sites Starting with 'CCA':** Filtered records where the launch site name begins with 'CCA', displaying the first 5 results.
- **Total Payload Mass by NASA (CRS):** Calculated the total payload mass carried by boosters for NASA (CRS) missions.
- **Average Payload Mass by Booster Version F9 v1.1:** Calculated the average payload mass for the booster version F9 v1.1.
- **First Successful Ground Pad Landing:** Identified the date of the first successful landing on a ground pad.
- **Boosters with Successful Drone Ship Landings:** Listed boosters that successfully landed on drone ships with a payload mass between 4000 and 6000 kg.

EDA with SQL

- **Mission Outcome Counts:** Summarized the total number of successful and failed mission outcomes.
- **Boosters with Maximum Payload:** Retrieved booster versions that carried the maximum payload mass.
- **Failed Drone Ship Landings in 2015:** Filtered records to display failed landings on drone ships in 2015, grouped by month.
- **Landing Outcome Rankings:** Ranked the count of different landing outcomes between June 2010 and March 2017 in descending order.

The GitHub URL of my completed EDA with SQL notebook:

[EDA with SQL](#)

Build an Interactive Map with Folium

Map Object 1: Markers

- Added markers to represent SpaceX launch sites, highlighting their geographic locations.
- **Purpose:** To allow users to quickly identify each launch site's coordinates on the map.

Map Object 2: Circles

- Circles were added to show proximity to various geographical features (e.g., closest cities, highways, railways).
- **Purpose:** To visualize distances and show connections between launch sites and key infrastructure.

Map Object 3: Lines

- Lines connecting launch sites with nearby features like highways or cities were drawn to represent physical connectivity.
- **Purpose:** To provide a visual representation of how launch sites are linked to transportation or urban areas.

Build an Interactive Map with Folium

Map Object 4: Popups

- Integrated popups displaying details about each launch site when clicking on a marker.
- **Purpose:** To give users immediate information about the site, such as the name and launch count, enhancing the interactive experience.

Other features:

- Added MousePosition to display live coordinates while hovering over the map.
- **Purpose:** To offer an interactive way to explore the map, including identifying specific locations.

The GitHub URL of my completed interactive map with Folium map:

[Interactive Map with Folium](#)

Build a Dashboard with Plotly Dash

Plot 1: Pie Chart

- A pie chart displaying the success/failure ratio of all SpaceX launches across different sites.
- **Purpose:** To provide an overview of the launch outcomes across all sites, allowing users to quickly understand the proportion of successful vs. failed launches.
- **Interaction:** A dropdown allows users to filter and view data for specific launch sites.

Plot 2: Scatter Plot

- A scatter plot showing the correlation between payload mass and launch success for different SpaceX sites.
- **Purpose:** To visualize the relationship between the payload mass and whether the launch was successful, giving insights into the performance of launches with different payload sizes.
- **Interaction:** A slider allows users to filter by payload mass range, and a dropdown enables filtering by launch site.

The GitHub URL of my completed Plotly Dash lab: [Plotly Dash](#)

Predictive Analysis (Classification)

Data Preprocessing:

- Collected and cleaned SpaceX mission data, preparing features like launch site, payload mass, and booster version for model training.
- Managed missing values and transformed categorical data into numerical form (e.g., using OneHotEncoding for categorical variables).

Model Selection:

- Implemented several classification algorithms:
 - Logistic Regression
 - Support Vector Machine (SVM)
 - Decision Trees
 - K-Nearest Neighbors (KNN)
- Split the dataset into training and test sets (80/20) to ensure robust model evaluation.

Predictive Analysis (Classification)

Model Evaluation:

- Used accuracy score, precision, recall, and F1-score to evaluate model performance.
- Conducted cross-validation for model robustness, ensuring consistent performance across different data splits.

Hyperparameter Tuning:

- Utilized GridSearchCV to optimize hyperparameters for each model, searching for the best parameter combinations to improve accuracy and performance.

Best Performing Model:

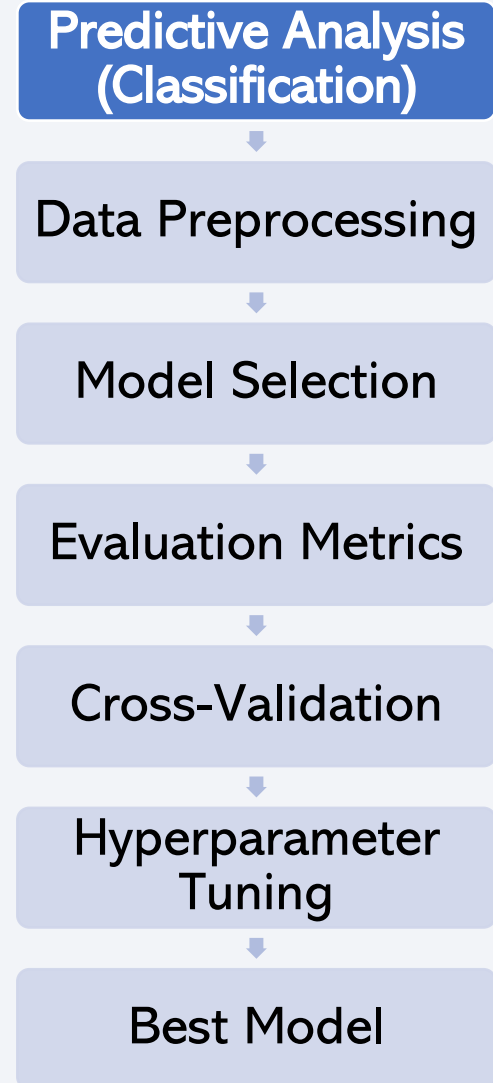
- Compared the models using test set performance and confusion matrix analysis.
- Found that [Best Model] had the highest accuracy and best trade-off between precision and recall.

Predictive Analysis (Classification)

The GitHub URL of my completed
predictive analysis lab:

[Predictive Analysis \(Classification\)](#)

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Results

Exploratory Data Analysis (EDA) Results:

Flight Number vs. Launch Site:

- The graph shows the relationship between the flight number and the launch site.
- Color coding distinguishes between successful (class 1) and unsuccessful (class 0) landings.
- Insights show certain launch sites like CCAFS SLC 40 have more frequent launches and generally higher success rates as flight numbers increase.

Payload Mass vs. Launch Site:

- This analysis illustrates the payload mass in kilograms against different launch sites.
- Payloads varied across sites, with larger masses (up to 16000 kg) seen in KSC LC 39A.
- Success rates are mapped using the class labels, providing insight into launch site performance based on payload.

Results

Exploratory Data Analysis (EDA) Results:

Success Rate vs. Orbit Type:

- The bar chart compares the success rate of missions based on orbit type.
- Some orbits, like GEO and ES-L1, show a 100% success rate, while others like GTO have a lower success rate.
- The analysis reveals which orbit types yield the most consistent mission outcomes.

Results

Interactive analytics demo in screenshots

SpaceX Launch Records Dashboard

All Sites

All Sites

CCAFS LC-40

VAFB SLC-4E

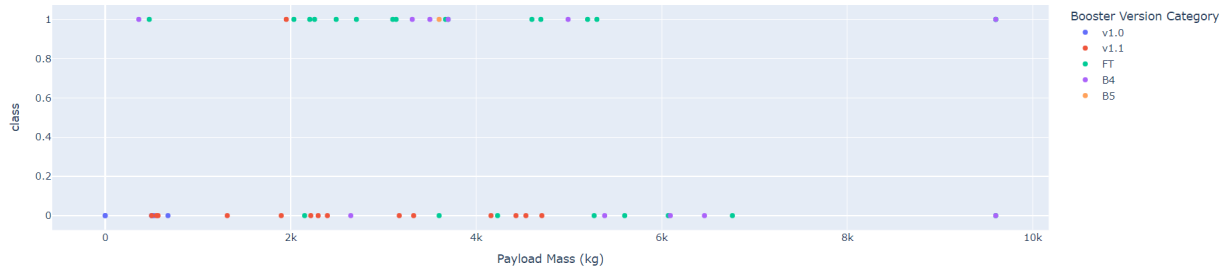
KSC LC-39A

CCAFS SLC-40

Payload range (Kg):



Correlation between Payload and Success for all Sites



Payload range (Kg):



SpaceX Launch Records Dashboard

All Sites

Total Success Launches by Site



SpaceX Launch Records Dashboard

CCAFS LC-40

Total Success Launches for site CCAFS LC-40



Results

Predictive analysis results

Logistic Regression Model (with GridSearchCV):

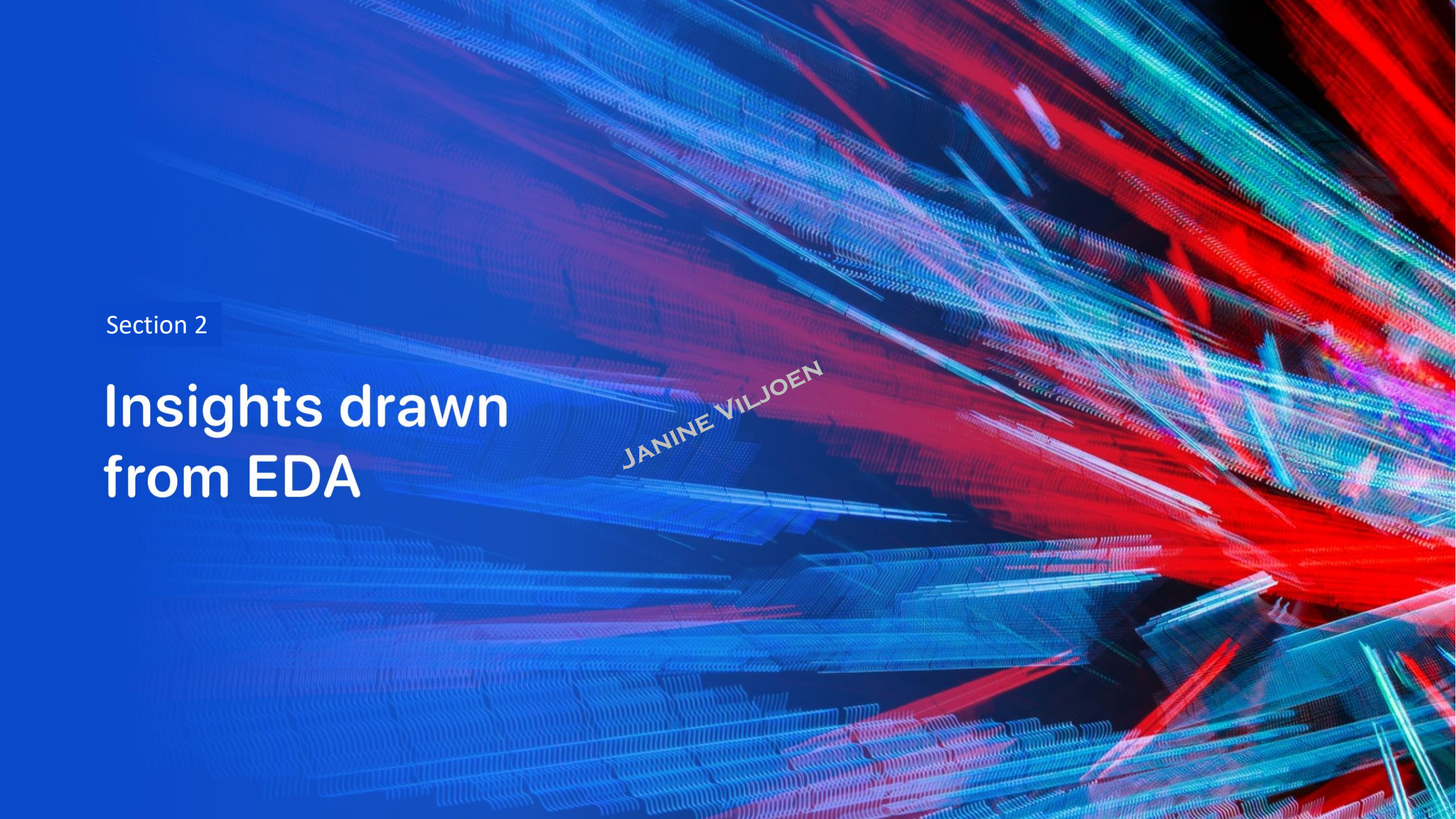
- The best parameters for Logistic Regression were identified as C: 0.01, penalty: l2, and solver: lbfgs with an accuracy of 84.6% on the validation set.
- On test data, it achieved 83.3% accuracy.
- A confusion matrix was used to evaluate performance, showing 12 true positives and 3 false positives.

Model Comparison:

- Multiple models, including Logistic Regression, SVM, Decision Tree, and KNN, were evaluated.
- All models performed with 83.3% accuracy, but Logistic Regression was selected as the best based on its consistent performance.

Final Model:

- Logistic Regression was selected as the final model due to its performance and simplicity, ensuring a balance between prediction accuracy and computational efficiency.

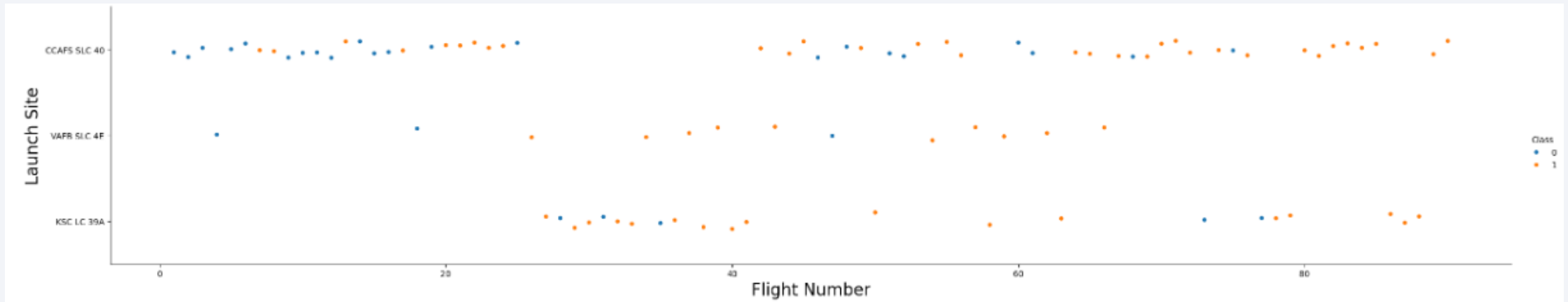


Section 2

Insights drawn from EDA

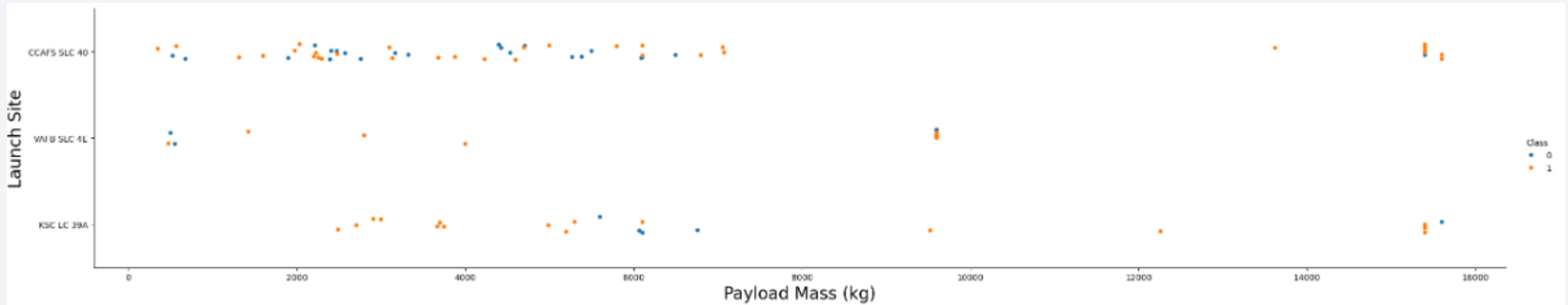
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Flight Number vs. Launch Site



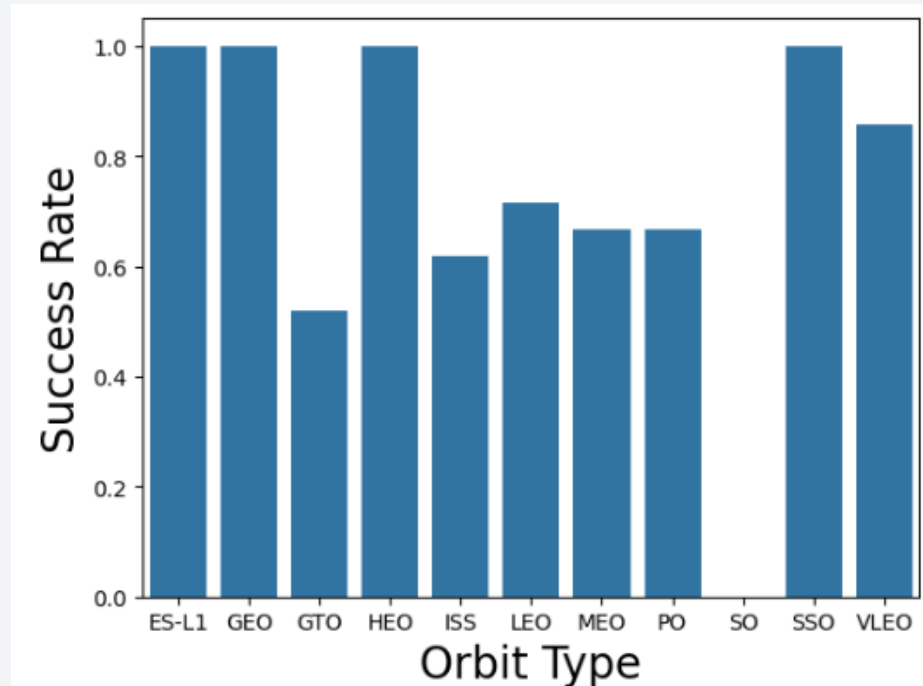
- The graph shows the relationship between the flight number and the launch site.
- Color coding distinguishes between successful (class 1) and unsuccessful (class 0) landings.
- Insights show certain launch sites like CCAFS SLC 40 have more frequent launches and generally higher success rates as flight numbers increase.

Payload vs. Launch Site



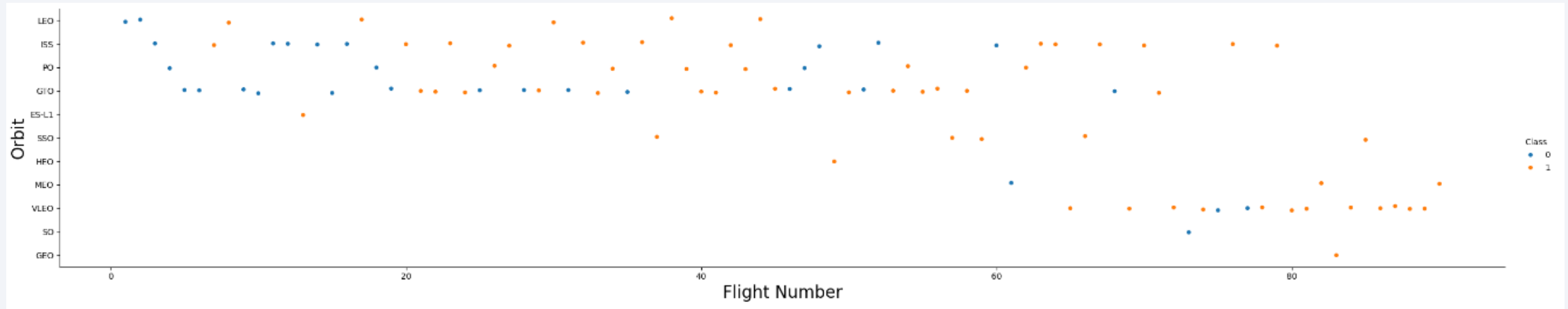
- This analysis illustrates the payload mass in kilograms against different launch sites.
- Payloads varied across sites, with larger masses (up to 16000 kg) seen in KSC LC 39A.
- Success rates are mapped using the class labels, providing insight into launch site performance based on payload.
- Now if you observe Payload Mass Vs. Launch Site scatter point chart you will find for the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than 10000).

Success Rate vs. Orbit Type



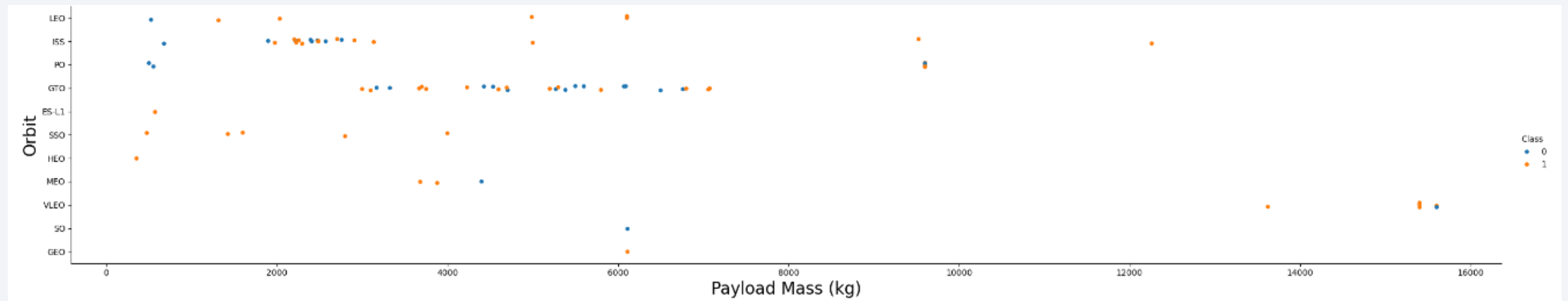
- The bar chart compares the success rate of missions based on orbit type.
- Some orbits, like GEO and ES-L1, show a 100% success rate, while others like GTO have a lower success rate.
- The analysis reveals which orbit types yield the most consistent mission outcomes.

Flight Number vs. Orbit Type



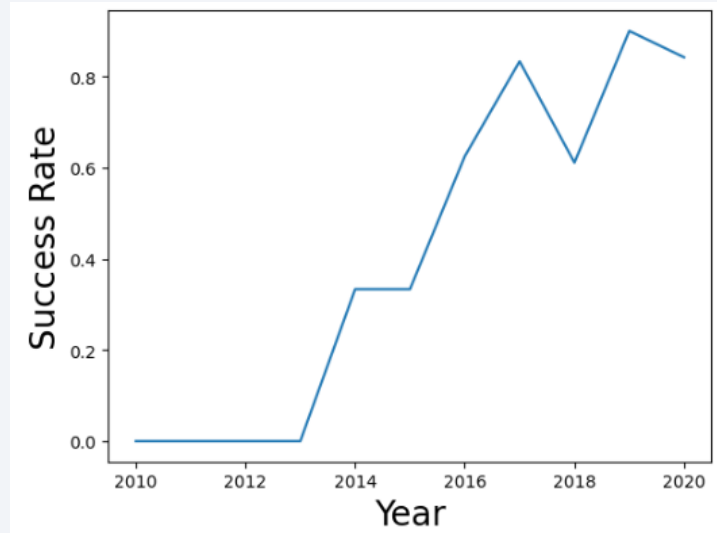
- You can observe that in the LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.

Payload vs. Orbit Type



- With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS.
- However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.

Launch Success Yearly Trend



- You can observe that the success rate since 2013 kept increasing till 2020.
- Early Years (Pre-2015): The success rate in the earlier years was lower, with more unsuccessful launches, reflecting the experimental nature of early launches.
- Post-2015: There was a clear upward trend in successful launches, indicating improvements in technology, processes, and experience.
- Recent Years (2020 and beyond): The trend showcases a near-perfect success rate, demonstrating SpaceX's operational maturity.

All Launch Site Names

Unique launch sites:

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

SQL Query:

```
%sql SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE;
```

Explanation:

- The query uses the DISTINCT keyword to retrieve unique values from the Launch_Site column in the SPACEXTABLE database.
- The result filters out any duplicate launch site entries and returns only the unique names.

Launch Site Names Begin with 'CCA'

5 records where launch sites begin with `CCA`:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

SQL Query: %sql SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;

Explanation:

- The query retrieves records from the SPACEXTABLE where the Launch_Site column begins with "CCA".
- The LIKE 'CCA%' clause filters launch sites that start with "CCA", which corresponds to Cape Canaveral Air Force Station (CCAFS) launch sites.
- The LIMIT 5 clause restricts the result to only 5 records to avoid retrieving the entire dataset.

Total Payload Mass

Total payload carried by boosters from NASA:

total_payload_mass
None

SQL Query:

```
%sql SELECT SUM("Payload_Mass__kg_") AS total_payload_mass FROM SPACEXTABLE WHERE  
"Mission_Outcome" LIKE '%NASA (CRS)%';
```

Explanation:

- The query attempts to calculate the total payload mass carried by boosters for NASA (CRS) missions by summing the Payload_Mass__kg_ field in the SPACEXTABLE.
- The LIKE '%NASA (CRS)%' clause filters the rows to include only those where the mission outcome includes NASA's Commercial Resupply Services (CRS) missions.

Average Payload Mass by F9 v1.1

The average payload mass carried by booster version F9 v1.1:

avg_payload_mass
2928.4

SQL Query:

```
%%sql SELECT AVG("Payload_Mass__kg_") AS avg_payload_mass FROM SPACEXTABLE WHERE  
"Booster_Version" = 'F9 v1.1';
```

Explanation:

- This query calculates the average payload mass for the booster version 'F9 v1.1' by taking the average of the Payload_Mass__kg_ column from the SPACEXTABLE where the Booster_Version is exactly 'F9 v1.1'.

First Successful Ground Landing Date

The dates of the first successful landing outcome on ground pad:

MIN(Date)
2015-12-22

SQL Query:

```
%%sql SELECT MIN(Date) FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)';
```

Explanation:

- This query retrieves the earliest date (MIN(Date)) from the SPACEXTABLE where the Landing_Outcome is 'Success (ground pad)'.

Successful Drone Ship Landing with Payload between 4000 and 6000

The names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000:

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

SQL Query:

```
%%sql SELECT "Booster_Version" FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (drone ship)' AND "Payload_Mass__kg_" BETWEEN 4000 AND 6000;
```

Explanation:

- This query fetches the Booster_Version from the SPACEXTABLE where the landing was successful on a drone ship, and the payload mass is between 4000 kg and 6000 kg.

Total Number of Successful and Failure Mission Outcomes

Total number of successful and failure mission outcomes:

Mission_Outcome	total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

SQL Query:

```
%%sql SELECT "Mission_Outcome", COUNT(*) AS total FROM SPACEXTABLE GROUP BY "Mission_Outcome";
```

Explanation:

- The query groups the data by the column Mission_Outcome and counts the number of occurrences for each outcome. This allows us to see how many missions were successful, how many failed, and any other distinct outcomes.

Boosters Carried Maximum Payload

The names of the booster which have carried the maximum payload mass:

SQL Query:

```
%%sqlSELECT "Booster_Version" FROM SPACEXTABLE WHERE  
"Payload_Mass__kg_" = (SELECT MAX("Payload_Mass__kg_") FROM  
SPACEXTABLE);
```

Explanation:

This query retrieves the names of boosters that carried the maximum payload mass. It uses a subquery to first identify the maximum payload mass in the dataset and then fetches the booster versions that carried this maximum payload.

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

The failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015:

month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

SQL Query:

```
%%sqlSELECT SUBSTR(Date, 6, 2) AS month, "Landing_Outcome", "Booster_Version",  
"Launch_Site" FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Failure (drone ship)' AND  
SUBSTR(Date, 1, 4) = '2015';
```

Explanation:

- This query extracts the month, Landing_Outcome, Booster_Version, and Launch_Site for failed drone ship landings in the year 2015.
- The SUBSTR(Date, 6, 2) function extracts the month, while SUBSTR(Date, 1, 4) ensures the data is filtered for the year 2015.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

The count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order:

SQL Query:

```
%%sqlSELECT "Landing_Outcome", COUNT(*) AS outcome_count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY "Landing_Outcome" ORDER BY outcome_count DESC;
```

Landing_Outcome	outcome_count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Explanation:

- The query counts the occurrences of each Landing_Outcome between the specified date range (June 4, 2010, to March 20, 2017).
- The results are grouped by Landing_Outcome and ordered in descending order based on the count (outcome_count), showing the most frequent outcomes first.

Section 3

Launch Sites Proximities Analysis

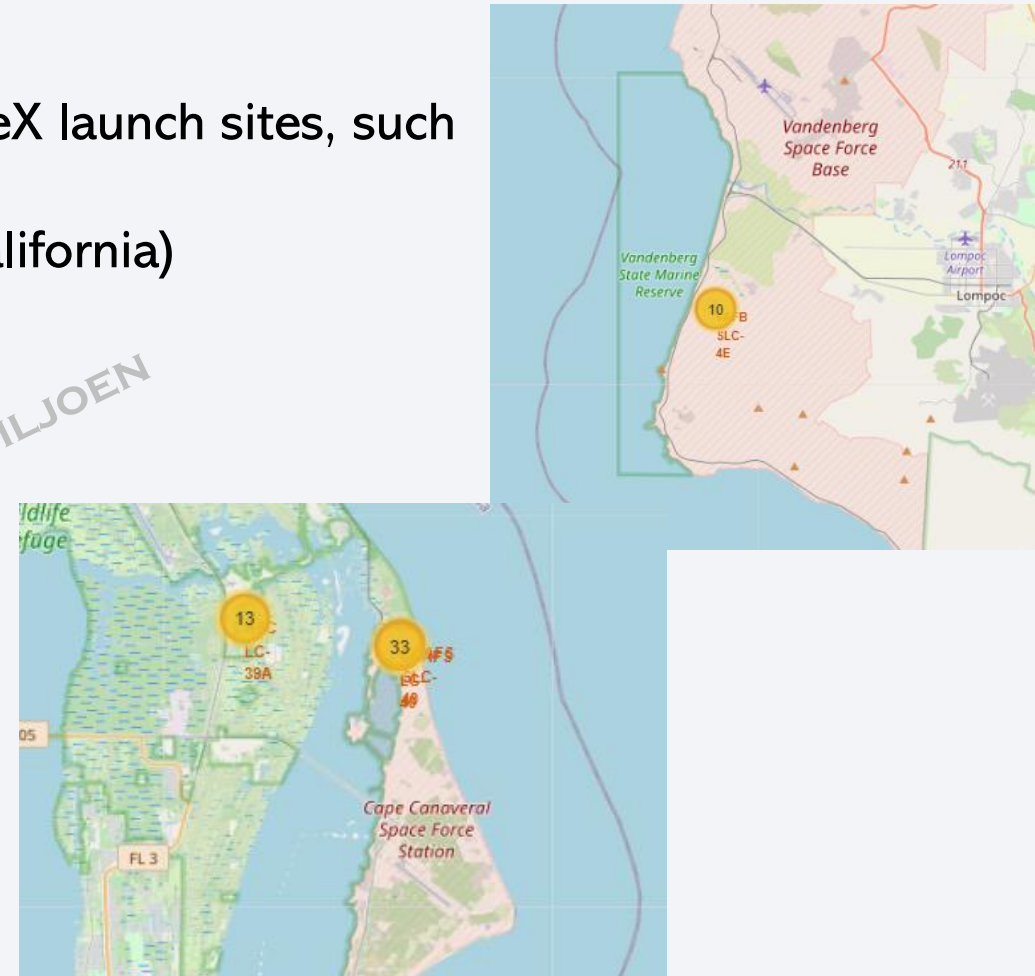
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Folium Map: Global Overview of SpaceX Launch Sites with Location Markers

Key Elements and Findings on the Map:

Highlighted Launch Sites: The map displays SpaceX launch sites, such as:

- VAFB SLC-4E (Vandenberg Air Force Base, California)
- KSC LC-39A (Kennedy Space Center, Florida)
- CCAFS LC-40 (Cape Canaveral, Florida)



Folium Map: Global Overview of SpaceX Launch Sites with Location Markers

Circle Markers: Each launch site is represented by a circle with a popup label showing the name of the site. The color and size (radius) of the circle emphasize the location's significance.

Custom Markers: Additionally, text markers are used to label each launch site, ensuring easy identification.

Geographical Spread: The map demonstrates that SpaceX launch sites are predominantly located in the U.S., primarily on the east coast in Florida and the west coast in California. These coastal locations provide ideal conditions for launching rockets over the ocean, minimizing risk to populated areas.

This visualization is crucial for understanding the geographical distribution of SpaceX's launch infrastructure, which is strategically positioned to support different types of missions, including polar and geostationary orbits.



Folium Map: SpaceX Launch Outcomes: Color-Labeled Map Representation

Key Elements and Findings on the Map:

Launch Outcomes by Color: Each marker on the map represents a SpaceX launch site, color-coded based on the launch outcomes (e.g., success or failure):

- Green: Successful landings.
- Red: Failed landings.

Interactive Popups: Clicking on each marker reveals more details about the launch, such as the mission name, outcome, and other specifics.

Global Distribution of Outcomes: By visualizing the outcomes directly on the map, it's easier to see which launch sites have had more successful missions and which ones have had challenges.

Importance of Coastal Locations: As with the earlier map, this one underscores the use of coastal sites for safety reasons, but now adds a layer of operational performance, illustrating the reliability of various launch sites over time.



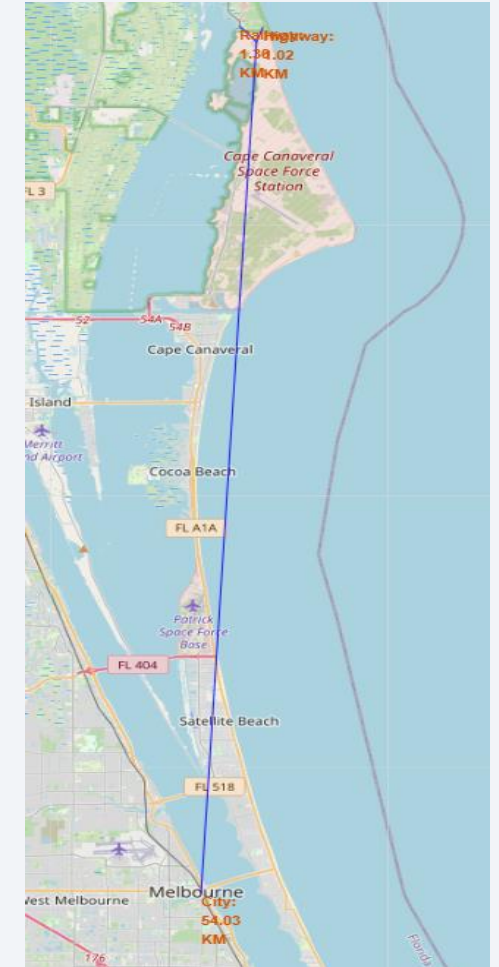
Folium Map: Launch Site Proximities: Railway, Highway, and Coastline Distances

Key Elements and Findings on the Map:

Selected Launch Site: The map focuses on a specific SpaceX launch site (e.g., CCAFS LC-40 or VAFB SLC-4E), with the location clearly marked by a circle or marker.

Nearby Infrastructure: The map highlights key proximities to the launch site:

- **Railway Proximity:** A marker or line indicating the nearest railway station, with the distance shown in kilometers or miles.
- **Highway Proximity:** A marker or line showing the nearest major highway or road, with its distance to the launch site.
- **Coastline Proximity:** The map shows the coastline, which is important for sea-based recovery missions, with the distance indicated.



Folium Map: Launch Site Proximities: Railway, Highway, and Coastline Distances

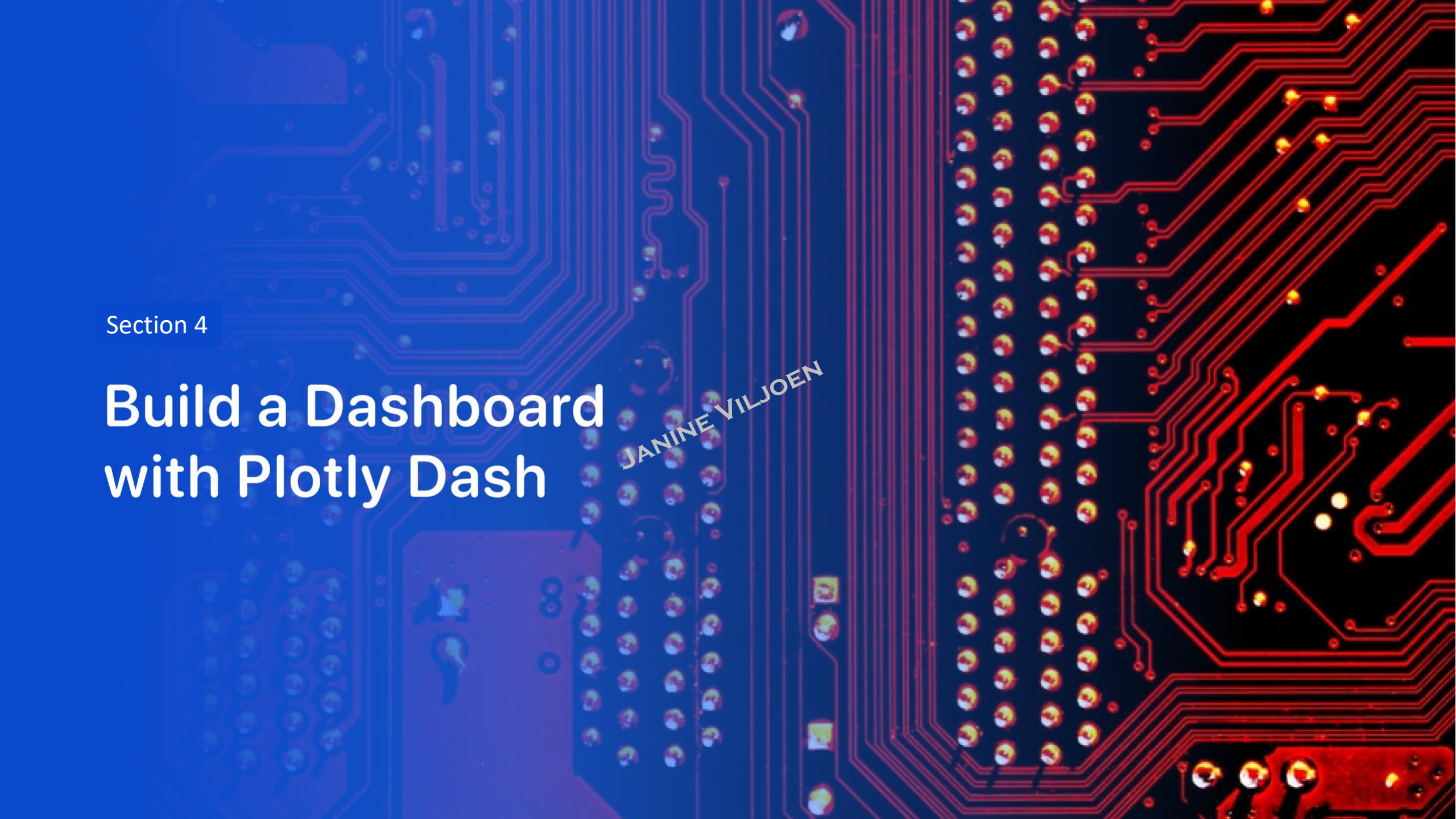
Distance Calculations: Each proximity (railway, highway, and coastline) has a popup or label showing the exact distance from the launch site to the respective infrastructure in real-time. This can be calculated using geopy or similar tools to measure distances between two geographical coordinates.

Important Findings:

- **Strategic Location:** Launch sites tend to be near coastlines for safety and logistical reasons, as rockets can be launched over water to minimize risks to populated areas.
- **Accessibility:** The proximity to highways and railways is crucial for transporting rockets, equipment, and personnel to the launch site.
- **Proximity to Railways:** If a site is relatively far from a railway, it might rely more on road transport, which may affect logistics and operational costs.

This visualization allows for the assessment of how well-connected the launch site is to major infrastructure, which can influence operational efficiency.





Section 4

Build a Dashboard with Plotly Dash

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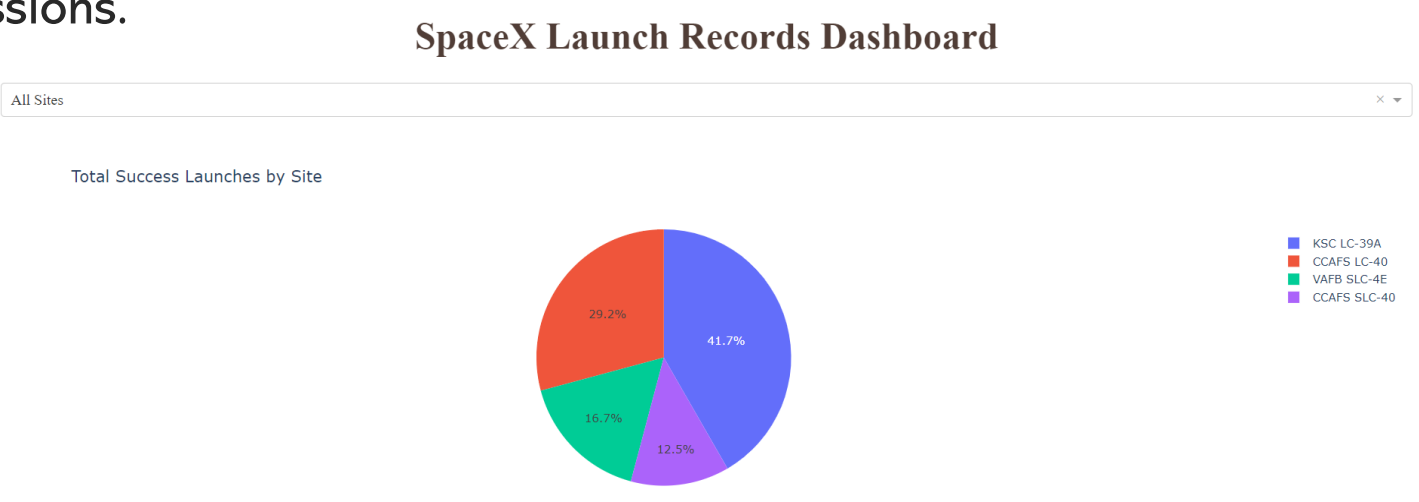
Launch Success Count for All Sites: Pie Chart Overview

Key Elements and Findings on the Screenshot:

Pie Chart Representation: The pie chart visualizes the distribution of successful launches across different launch sites. Each segment of the pie corresponds to a specific launch site, with the size of the segment proportional to the number of successful launches at that site.

Launch Site Labels: The chart includes labels or a legend identifying the different SpaceX launch sites, such as CCAFS LC-40, KSC LC-39A, VAFB SLC-4E, and others. Each site is assigned a unique color for easy differentiation.

Important Findings: The pie chart reveals which launch site has the highest number of successful launches. Typically, CCAFS LC-40 or KSC LC-39A might be leading, reflecting their frequent use for major SpaceX missions.



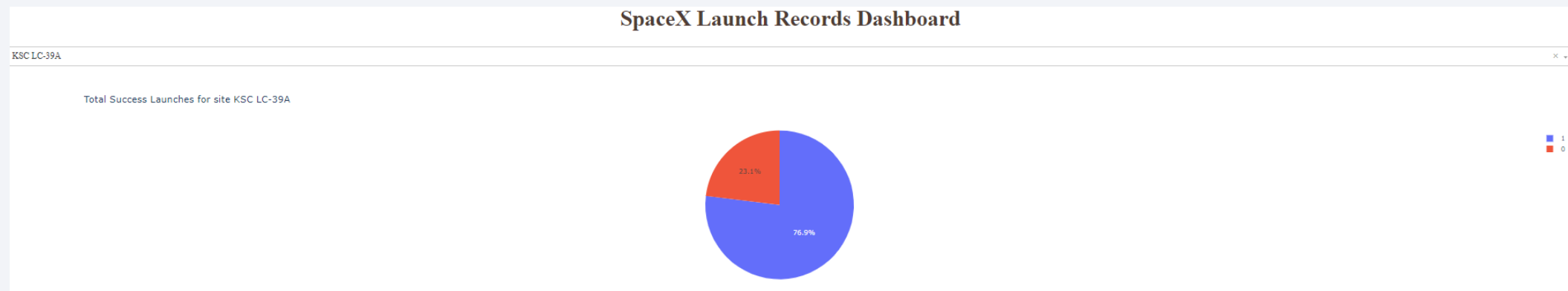
Launch Success Ratio: Pie Chart for Launch Site with Highest Success Rate

Key Elements and Findings on the Screenshot:

This pie chart focuses exclusively on the launch site with the highest success ratio, displaying the proportion of successful versus failed launches at that specific site. The pie chart shows two key outcomes: Success: The portion of the chart that represents the number of successful launches at the selected launch site. Failure: The smaller portion reflects the failed launch attempts at the same site.

Important Findings:

The chart shows that the majority of launches at this site have been successful, making this site crucial to SpaceX's overall launch success. Minimal failures are visible, they highlighting the reliability and operational efficiency of the site.



Correlation between Payload and Launch Outcome for All Sites

This scatter plot demonstrates the correlation between payload mass and launch outcomes for different booster versions across all launch sites. Here's an explanation of key elements and findings:

Payload Range Slider (Top):

This slider allows you to filter launches based on their payload mass (kg), ranging from 0 to 10,000 kg. By adjusting this slider, you can observe how different payload ranges affect the success or failure of launches.

Payload range (Kg):



Correlation between Payload and Launch Outcome for All Sites

X-Axis (Payload Mass (kg)): Represents the mass of the payload for each launch. The scale ranges from 0 to 10,000 kg, and data points show launches with varying payloads.

Y-Axis (Class): Represents the launch outcome, where 1 indicates success and 0 indicates failure.

Color-Coded Booster Versions: The scatter plot uses different colors to represent various booster versions: **v1.0 (Blue):** Early version of the booster. **v1.1 (Red):** An updated version. **FT (Green):** Full Thrust variant, used more frequently in successful launches. **B4 (Purple):** Block 4 version, leading to some successful launches. **B5 (Orange):** The latest version with the highest success rate.



Correlation between Payload and Launch Outcome for All Sites

Key Findings:

Success Rate by Payload Range:

- Payloads Under 6,000 kg: Most launches with payloads under 6,000 kg tend to be successful (class = 1), particularly those using the FT and B4 boosters (green and purple dots).
- Payloads Over 6,000 kg: There are fewer launches with payloads exceeding 6,000 kg, and the success rate becomes more variable. However, B5 (orange) appears more successful at handling higher payloads.

Booster Version with Highest Success Rate:

- FT (Full Thrust, Green): Most successful across various payload ranges, especially under 6,000 kg.
- B5 (Block 5, Orange): Newest and most reliable version for larger payloads, showing a good success rate in higher ranges.

Booster Versions with Lower Success Rates:

- v1.0 (Blue) and v1.1 (Red): These earlier versions have more failed launches, particularly with payloads below 2,000 kg.

Effect of Payload Mass on Outcome:

- The larger the payload, the more challenging it appears to achieve successful launches. The FT and B5 boosters show better performance with heavier payloads, while earlier versions like v1.0 and v1.1 struggle even with smaller payloads.

Section 5

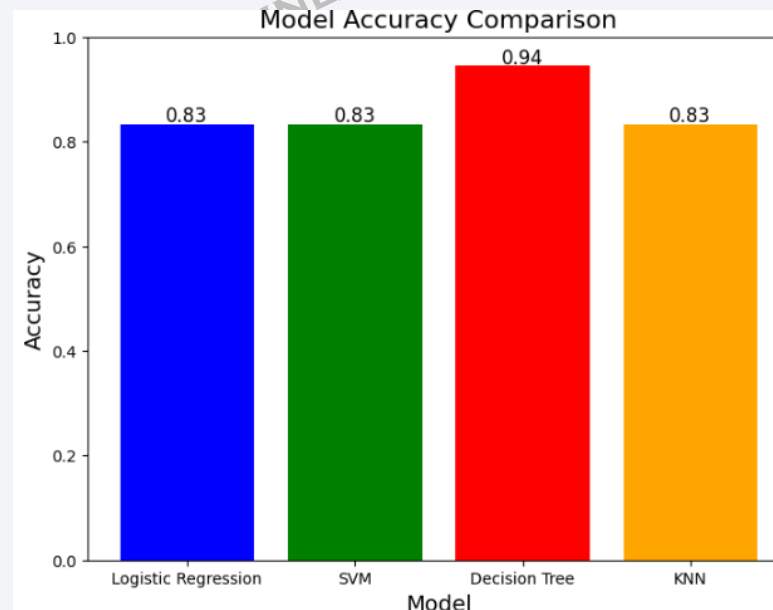
Predictive Analysis (Classification)

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Classification Accuracy

Model with Highest Accuracy:

- The Decision Tree model has the highest accuracy, with a score of 0.9444, outperforming the other models. The **Decision Tree** model is the most accurate in predicting the SpaceX first-stage landing outcomes in this scenario.
- Other Models: Logistic Regression, SVM, and KNN models all have similar accuracies of 0.8333, indicating that these models performed similarly but slightly lower than the Decision Tree.



Confusion Matrix: Best Performing Model - Decision Tree

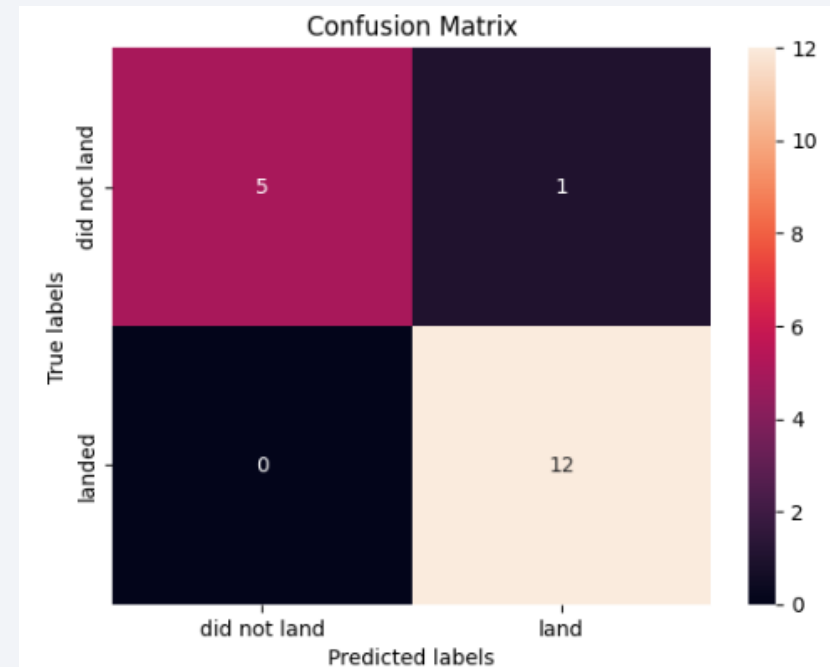
True Positives (TP) = 12: The model correctly predicted 12 successful landings, which is an excellent result, indicating high precision for success predictions.

True Negatives (TN) = 5: The model correctly identified 5 instances where the landing did not succeed, showing strong performance in identifying failures.

False Positives (FP) = 1: There was only one case where the model incorrectly predicted a landing success, but it was actually a failure. This low number of FP indicates that the model makes very few incorrect positive predictions.

False Negatives (FN) = 0: There were no instances where the model incorrectly predicted a failure when it should have predicted a success, which is an ideal outcome.

Conclusion: The Decision Tree model performs very well, with high accuracy and minimal errors in classifying landings. The absence of false negatives (0 FN) and only one false positive demonstrates the model's capability to distinguish between successful and unsuccessful landings with high reliability. This is reflected in the high accuracy score (0.9444).



Conclusions

- This project effectively demonstrated the use of historical SpaceX Falcon 9 launch data to predict the success of rocket landings.
- By employing a combination of data collection, cleaning, and analysis methods—such as API calls, web scraping, and exploratory data analysis—the team was able to uncover key insights about launch characteristics.
- Predictive modeling techniques, including logistic regression, SVM, and decision trees, were evaluated, with the decision tree model emerging as the best performer, achieving an accuracy of 94.4%.
- These findings provide a foundation for SpaceX to enhance its landing success rates, contributing to cost reduction and mission efficiency improvements.

Appendix

- The GitHub URL of my completed SpaceX API calls notebook: [GitHub URL SpaceX API Calls](#)
- The GitHub URL of my completed web scraping notebook: [GitHub URL Webscraping](#)
- The GitHub URL of my completed data wrangling related notebooks: [GitHub URL Data Wrangling](#)
- The GitHub URL of my completed EDA with data visualization notebook: [GitHub URL EDA with Data Visualization](#)
- The GitHub URL of my completed EDA with SQL notebook: [EDA with SQL](#)
- The GitHub URL of my completed interactive map with Folium map: [Interactive Map with Folium](#)
- The GitHub URL of my completed Plotly Dash lab: [Plotly Dash](#)
- The GitHub URL of my completed predictive analysis lab: [Predictive Analysis \(Classification\)](#)

Thank you!

JANINE VILJOEN

